MILITARY SPECIFICATION

DETONATOR, ELECTRIC, T44
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers loading, assembling, and packing for one type of detonator, designated as Detonator, Electric, T44.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-14077 - Polytetrafluoroethylene (Teflon) Molding and Extruding Material.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

8838102 - Detonators Electric, T44, Assembly and Details.
8798699 - Box, Packing, Ammunition, for Detonators, Electric, Assembly, Details and Packing.
8798698 - Box, Set-Up, Packing, Ammunition, For Detonators, Electric, Assembly, Details, Packing and Marking.
8800784 - Ammunition Container, Metal, For Detonators, Electric, Assembly, Details, Packing, and Marking.

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PUBLICATIONS

ORDNANCE INSPECTION HANDBOOK

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-8781494 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable specifications and drawings.

3.2 Assembly.-The detonator assemblies shall comply with all requirements specified on Drawing (dwg.) 8838102, and with all requirements specified in applicable specifications.

3.3 Lead azide moisture content.-The moisture content of the lead azide at the loading station, at the time of loading the detonators, shall not exceed 0.50 percent when determined as specified in 4.3.1.

3.4 PETN moisture content.-The moisture content of the PETN at the loading station, at the time of loading the detonators, shall not exceed 0.50 percent when determined as specified in 4.3.2.

3.5 Waterproofness.-The detonator assemblies shall be waterproof, as evidenced by normal functioning after immersion in water, when tested as specified in 4.3.3.

3.6 Functioning.

3.6.1 Time.-The detonators shall fire on the first attempt within 100 microseconds when energy from a 16 microfarad capacitor, maximum (max.), is applied as specified in 4.3.3.1.1.

3.6.2 Output.-The detonators shall fire on the first attempt when energy from a 16 microfarad capacitor, max., charged to a 2.5 volts, max., is applied as specified in 4.3.3.1.2 and shall cause perforation of the lead disc to not less than 3/16 inch diameter, minimum (min.).

3.6.3 Duds.-The detonators shall fire on the first attempt when energy from a 16 microfarad capacitor, max., charged to 2.5 volts, max., is applied as specified in 4.3.3.1.3.

3.7 Workmanship.-All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring, and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

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d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings, and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor

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immediately. To avoid interference with operations, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of detonators loaded and assembled by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government, shall not be cause for changing the lot number. Each lot shall contain:

a. Metal parts of one interfix lot number from one manufacturer.
b. Lead azide of one lot from one manufacturer.
c. PETN of one lot from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Bridge and plug assembly, prior to assembly of spot charge (see dwg. 8838108).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
	101. Bridge missing	Visual
Minor:	None defined	

4.2.2.2 Bridge and plug assembly (see dwg. 8838108).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
	101. Spot charge missing or cracked	Visual
Minor:	AQL 0.40 percent	
	201. Evidence of poor workmanship (see 3.7)	Visual

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4.2.2.3 Assembly (see dwg. 8838102)

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Total length, max.	Gage
102.	Diameter, max.	Gage
Minor:	AQL 2.50 percent	
201.	Length to end of cup, max.	Gage
202.	Crimp not a full 360°	Visual
203.	Contact surface of pin not clean	Visual
204.	Varnish seal missing from cup crimp	Visual
205.	Cup contains deep scratches or grooves	Visual
206.	Marking misleading or unidentifiable	Visual
207.	Evidence of poor workmanship (see 3.7)	Visual

4.2.2.4 Sealed metal container (see dwg. 8800784).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Contents exposed	Visual
102.	Sealing strip incomplete or badly wrinkled	Visual
Minor:	AQL 1.50 percent	
201.	Contents move when shaken	Manual
202.	Marking misleading or unidentifiable	Visual
203.	Bare spot on exterior other than slight scratches	Visual

4.2.2.5 Box, set-up, packing (see dwg. 8798698).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Waterproof seal broken by puncture, cut or tear	Visual
102.	Sealing strip incomplete or badly wrinkled	Visual

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Minor:	AQL 1.00 percent	
201.	Contents move when shaken	Manual
202.	Marking misleading or unidentifiable	Visual

4.2.2.6 Sealed wood packing box (see dwg. 8798699).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Strapping missing, broken or loose	Visual-Manual
102.	Contents exposed	Visual
103.	Board broken or split	Visual
104.	DOD symbol misleading or unidentifiable	Visual
Minor:	None defined	

4.2.3 Testing.

4.2.3.1 Moisture content of lead azide (see 3.3) and moisture content of PETN (see 3.4) - Major defect.-A sample of 2 grams of lead azide and 5 grams of PETN shall be selected from that quantity used to produce a lot of detonators, for determination of moisture content. The determination shall be performed as specified in 4.3.1 and 4.3.2 respectively.

4.2.3.2 Functioning and waterproofness.-A sample of 185 detonator assemblies shall be selected from each lot for the combined tests. The sequence of testing and the number of samples for individual tests, shall be as follows:

4.2.3.2.1 Functioning time and duds (see 3.6.1) - Major defect.-Thirty five detonator assemblies shall be subjected to the functioning time test specified in 4.3.3.1.1 using test equipment as required in 4.2.4. The lot shall be rejected if the mean functioning time ($\bar{X}$) plus the product of 2.27 and the standard deviation of the sample (s) exceeds 100 microseconds. Detonators that fail to fire shall not be considered in the calculations, but will be replaced with detonators from the basic sample. When three duds are found, the lot shall be rejected and subsequent acceptance tests for the Government shall not be performed. The standard deviation shall be calculated using the formula in 6.2 or equivalent.

4.2.3.2.2 Output and duds (see 3.6.2) - Major defect.-If the lot has met the requirements of 3.6.1, fifty detonator assemblies shall be tested as specified in 4.3.3.1.2 using test equipment as specified in 4.2.4. The lot shall be rejected if two or more detonators fail to produce a perforation in the disc of the required size. Detonators that fail to fire shall be replaced with detonators from the

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basic sample. If three or more duds are found when testing for time and output, the lot shall be rejected and further acceptance testing for the Government shall not be performed. If the number of duds found during the combined tests does not exceed two, and only one detonator fails to produce the required disc perforation, the balance of the detonator assemblies in the basic sample shall be tested as specified in 4.3.3.1.2. The lot shall be rejected and acceptance testing shall be discontinued if one additional detonator fails to produce the required disc perforation during the second test, or a total of three duds are found in combined tests.

4.2.3.2.3 Duds (see 3.6.3) - Major defect.-If the fifty detonator assemblies tested complied with the requirements of 3.6.2, the balance of the detonators in the basic sample shall be subjected to the test specified in 4.3.3.1.3 using test equipment in accordance with 4.2.4. The lot shall be rejected and acceptance testing discontinued when a total of three duds are found in combined tests. If two duds are found, a second sample of 370 detonator assemblies shall be selected and subjected to the test specified in 4.3.3.1.3. Acceptance testing shall be discontinued and the lot rejected whenever the number of duds totals three.

4.2.3.3 Resistance between pin and cup (see dwg. 8838102) - Major defect.- This test shall be performed 100 percent. Any assembly which fails to meet the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.4.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment No. 8781494 or approved equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Determination of moisture content of lead azide.-A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of teflon (see MIL-P-14077) 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed specimen of approximately two grams shall be transferred to the moisture

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dish while both the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95° plus or minus (±) 2 degree Centigrade (°C.) for 2 hours. The cover shall be placed in the sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{(A-B)}{A} \times 100$$

where:

A = the weight of the specimen before drying.

B = the weight of the specimen after drying.

4.3.2 Determination of moisture content of PETN.-Approximately 5 grams of PETN shall be placed in a tared glass weighing dish. The dish and its contents shall be weighed, dried at 100 ± 2°C. for 1 hour; cooled in a desiccator and weighed. The loss in weight shall be calculated to the percentage of moisture in the PETN.

4.3.3 Waterproofness and functioning.

4.3.3.1 Conditioning.-The detonator assemblies shall be submerged for not less than 48 hours at a depth of 2 to 3 inches in water at a temperature of 20 ± 1°C. The assemblies shall be removed from the water bath and subjected to the following tests within one hour.

4.3.3.1.1 Time and duds.-The detonators shall be connected to the applicable test circuit specified in 4.2.4 and fired by applying the energy from a 16 microfarad capacitor, max., charged to a max., of 2.5 volts. The test shall be conducted at a temperature of 70 ± 10° Fahrenheit (F°).

4.3.3.1.2 Output and duds.-The detonators, conditioned at a temperature of 70 ± 10°F. shall be assembled in the applicable test equipment specified in 4.2.4, and fired by applying the energy from a 16 microfarad capacitor, max., charged to a max., of 2.5 volts. The perforation in the lead disc shall be observed for compliance with the requirement.

4.3.3.1.3 Duds.-The detonators shall be fired by applying the energy from a 16 microfarad capacitor, max., charge to a max., of 2.5 volts.

4.3.4 Resistance between the pin and the cup.-The detonators shall be connected to the test equipment specified in 4.2.4, and the electrical resistance measured using not more than 1 milliampere of current. The test shall be conducted behind a suitable barricade.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The detonators shall be preserved and packaged in accordance with dwg. 8800784.

5.1.2 Intermediate packaging.-The detonators shall be packaged in accordance with dwg. 8798698.

5.2 Packing.

5.2.1 Level C.-The containers shall be packed in accordance with dwg. 8798699.

5.3 Marking.-Marking shall be in accordance with dwg. 8798698, 8798699, and 8800784, as applicable.

5.4 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Standard deviation.-The following formula, or its equivalent, shall be used to compute the sample standard deviation:

$$S = \sqrt{\frac{\sum(\bar{X}-x)^2}{n-1}}$$

where:

S = sample standard deviation
$\bar{X}$ = sample mean time
x = individual observed time
$\sum(\bar{X}-x)^2$ = the sum of the square of the sample mean time minus each individual observed time
n = sample size

6.3 Check test for possible deterioration.-The following tests are designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The tests should be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check tests be performed, the contracting officer should insert a provision for the performance of the check tests in the contract. Two hundred detonators shall be selected at random for these tests.

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6.3.1 Functioning.

6.3.1.1 Time and duds.-Forty (40) detonators shall be subjected to the test specified in 4.3.3.1.1. Observation shall be made for the functioning time and all duds that may occur will be recorded. Duds will be replaced by detonators from the 200 selected. If the mean time ($\bar{X}$) plus 2.03 times the standard deviation (s) exceeds 100 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in paragraph 6.2 or equivalent. Duds shall not enter into the calculation for standard deviation.

6.3.1.2 Output and duds.-Fifty (50) detonators shall be subjected to the test specified in 4.3.3.1.2. Observation shall be made for the output and the number of duds that may occur will be recorded. Duds will be replaced by detonators from the 200 selected. If two detonators fail to adequately perforate the lead disc, a retest shall be permitted using the balance of the 200 detonators selected. If during the retest one more detonator fails to adequately perforate the lead disc, the retest shall be stopped and the lot rejected. During the retest the number of duds that may occur will be recorded. If three or more detonators fail to adequately perforate the lead disc during the original and combined tests the lot will be rejected.

6.3.1.3 Duds.-If no retest is required in the output phase, the balance of the detonators selected shall be subjected to the test for duds as specified in 4.3.3.1.3. The number of duds that may occur will be replaced. If more than two but not more than six duds are found a retest will be permitted using double the number of detonators selected in accordance with 6.3. The lot shall be rejected during retest as soon as the total number of duds observed in 6.3.1.1, 6.3.1.2 and 6.3.1.3 totals seven. If in the original sample of 200, seven or more duds are observed, the lot shall be rejected without retest.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
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